Experiment 9: Property Data Analysis Using Pandas

Aim

To analyze property data using Pandas and find the average listing price in each location, the number of properties with more than four bedrooms, and the property with the largest area.

Algorithm

1. Import the Pandas library.
2. Create a DataFrame containing property IDs, locations, bedrooms, areas, and listing prices.
3. Use groupby() and mean() to calculate the average listing price for each location.
4. Filter properties with more than four bedrooms and count them using sum().
5. Use idxmax() to identify the property with the largest area.
6. Display all results.

Code

```
import pandas as pd

# Create property data

property_data = pd.DataFrame({

   "Property_ID": [101, 102, 103, 104, 105],

   "Location": [

      "Chennai", "Hyderabad", "Chennai",

      "Bengaluru", "Hyderabad"

   ],

   "Bedrooms": [3, 5, 4, 6, 5],

   "Area_sqft": [1500, 2500, 1800, 3000, 2200],

   "Listing_Price": [5000000, 8000000, 6000000,

                     9000000, 7500000]

})

# 1. Average listing price in each location

average_price = property_data.groupby(
```

```
    "Location"
)["Listing_Price"].mean()
print("Average listing price by location:")
print(average_price)
# 2. Count properties with more than 4 bedrooms
count_properties = (
    property_data["Bedrooms"] > 4
).sum()
print("\nProperties with more than 4 bedrooms:",
      count_properties)
# 3. Property with the largest area
largest_property = property_data.loc[
    property_data["Area_sqft"].idxmax()
]
print("\nProperty with the largest area:")
print(largest_property)
```

Input

Property_ID	Location	Bedrooms	Area_sqft	Listing_Price
101	Chennai	3	1500	5000000
102	Hyderabad	5	2500	8000000
103	Chennai	4	1800	6000000
104	Bengaluru	6	3000	9000000
105	Hyderabad	5	2200	7500000

Output

```
Average listing price by location:
Location
Bengaluru    9000000.0
```

Chennai 5500000.0

Hyderabad 7750000.0

Properties with more than 4 bedrooms: 3

Property with the largest area:

Property_ID 104

Location Bengaluru

Bedrooms 6

Area_sqft 3000

Listing_Price 9000000

```
====== RESTART: C:/Users/sivak/Desktop/FODS LAB experiments codes/exp 9.py =====
Average listing price by location:
Location
Bengaluru    9000000.0
Chennai      5500000.0
Hyderabad    7750000.0
Name: Listing_Price, dtype: float64

Properties with more than 4 bedrooms: 3

Property with the largest area:
Property_ID            104
Location         Bengaluru
Bedrooms                 6
Area_sqft             3000
Listing_Price      9000000
Name: 3, dtype: object
```

Result

Thus, the average listing price for each location was calculated, three properties with more than four bedrooms were identified, and Property ID 104 was found to have the largest area of 3,000 square feet.